



CLASSROOM CONTACT PROGRAMME

(Academic Session : 2021 - 2022)

JEE(Main)

UNIT TEST

09-05-2021

JEE(Main+Advanced) : ENTHUSIAST COURSE [PHASE : S, I(A), I, LIVE-I(A) & LIVE-I]

Do not open this Test Booklet until you are asked to do so.

Read carefully the Instructions on this Test Booklet.

PAPER-2

Paper : Physics, Chemistry & Mathematics

Important Instructions :

1. Immediately fill in the form number on this page of the Test Booklet with Blue/Black Ball Point Pen. Use of pencil is strictly prohibited.
2. The candidates should not write their Form Number anywhere else (except in the specified space) on the Test Booklet/Answer Sheet.
3. The test is of **3 hours** duration.
4. The Test Booklet consists of **90** questions. The maximum marks are **300**.
5. There are **three** parts in the question paper 1,2,3 consisting of **Physics, Chemistry and Mathematics** having **30 questions** in each subject and each subject having **Two sections**.
 - (i) Section-I contains 20 multiple choice questions with **only one correct** option.
Marking scheme : +4 for correct answer, 0 if not attempted and –1 in all other cases.
 - (ii) Section-II contains 10 **Numerical Value Type** questions. Attempt any 5 questions. First 5 attempted questions will be considered for marking.
Marking scheme : +4 for correct answer and 0 in all other cases.
6. Use **Blue/Black Ball Point Pen only** for writing particulars/markings responses on **Side-1** and **Side-2** of the Answer Sheet. **Use of pencil is strictly prohibited**.
7. No candidate is allowed to carry any textual material, printed or written, bits of papers, mobile phone any electronic device etc, except the Identity Card inside the examination hall/room.
8. Rough work is to be done on the space provided for this purpose in the Test Booklet only.
9. On completion of the test, the candidate must hand over the Answer Sheet to the invigilator on duty in the Room/Hall. **However, the candidate are allowed to take away this Test Booklet with them.**
10. **Do not fold or make any stray marks on the Answer Sheet.**

Name of the Candidate (in Capitals) _____

Form Number : in figures _____

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Candidate's Signature : _____ Invigilator's Signature : _____

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Topic :

PHYSICS

SEC-A : (Maximum Marks: 80)

This section contains **20 questions**. Each question has 4 options for correct answer. Multiple-Choice Questions (MCQs) **Only one option is correct**. For each question, marks will be awarded as follows:

Full Marks : +4 If correct answer is selected.

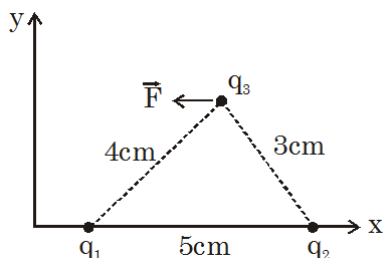
Zero Marks : 0 If none of the option is selected.

Negative Marks : -1 If wrong option is selected.

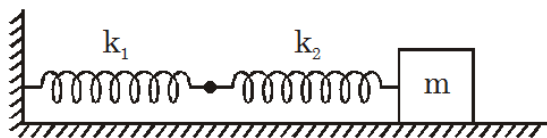
1. A lensmaker has to design an achromatic combination of effective power -0.8 D. The two types of glasses to be used (A and B) have dispersive powers in the ratio $2 : 3$. Then the lens made from glass B will have focal length :
 (A) 41.6 cm
 (B) 62.5 cm
 (C) -41.6 cm
 (D) -62.5 cm
2. A beam of diameter 'D' is incident on a combination of convex lens of focal length 20 cm and a concave lens of focal length 10cm, placed at distance x from each other. It is observed that emergent beam from the concave lens is parallel to their common principle axis and diameter 'd' then :
 (A) $d = \frac{D}{4}$
 (B) $d = \frac{D}{2}$
 (C) $x = 30$ cm
 (D) $x = 20$ cm

3. In an uniform electric field, the potential difference between two points separated by 10 cm is 2 volt. Magnitude of electric field
 (A) is 20 V/m.
 (B) is less than 20 V/m.
 (C) may be greater than 20 V/m.
 (D) may be less than 20 V/m.
4. A lens is placed between a source of light and a wall. It forms images of area 2.5 mm^2 and 10 mm^2 on the wall, for its two different positions. The area of the source of light is (source and wall)
 (A) 5 mm^2 (B) 10 mm^2
 (C) 8 mm^2 (D) 2 mm^2
5. A particle performs simple harmonic motion with amplitude A. If its speed is doubled at the instant when it is at distance $\frac{A}{3}$ from equilibrium position. The new amplitude of the motion is :
 (A) $\sqrt{11} A$ (B) $\frac{\sqrt{33} A}{3}$
 (C) $\frac{\sqrt{33} A}{2}$ (D) $\frac{\sqrt{11} A}{3}$
6. Two SHM are represented by equations,
 $y_1 = 6 \cos\left[6\pi t + \frac{\pi}{6}\right]$
 $y_2 = 3 (\sqrt{3} \sin 3\pi t + \cos 3\pi t)$
 (A) Ratio of their amplitudes is 1.
 (B) Ratio of their time period is 1.
 (C) Ratio of their maximum velocities is 1.
 (D) Ratio of their maximum acceleration is 1.

7. Three charges q_1 , q_2 and q_3 are placed as shown. The magnitude of q_1 , is $2\mu\text{C}$ but its sign and the value of q_2 are not known. Charge q_3 is $+4\mu\text{C}$, and the net force $\vec{F}$ on q_3 is in the negative x direction.



- (A) The magnitude of charge q_1 is $\frac{32}{27}\mu\text{C}$
- (B) Charge q_1 is positive.
- (C) Charge q_2 is positive.
- (D) The magnitude of net force $\vec{F}$ on charge is $\frac{45}{22}\text{mN}$
8. In the given figure block executes SHM with total energy of oscillations E . The ratio of k_1 and k_2 is $2 : 1$. The energy associated with spring of spring constant k_1 when block is at $\frac{A}{2}$ distance away from mean position is (A is the amplitude) :



- (A) $\frac{E}{12}$ (B) $\frac{E}{4}$
- (C) $\frac{E}{3}$ (D) $\frac{E}{6}$

9. A thin prism P_1 with angle 4° and made from a glass of refractive index 1.48 is combined with another prism P_2 made from crown glass of refractive index 1.32 to produce dispersion without deviation. The angle of the prism P_2 is

(A) 4.33° (B) 3° (C) 6° (D) 3.8°

10. A block of mass $m = 3\text{ kg}$ resting on a horizontal frictionless floor is horizontally struck by a 9 N force that acts for 0.02 sec . After 3 sec it receives a second blow of force 9 N but in opposite direction which acts for 0.01 sec . The speed of the body after 10 sec is

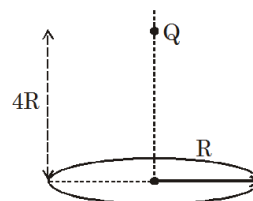
(A) 0 (B) 3 cm/sec

(C) 90 cm/sec (D) 30 cm/sec

11. Refractive index of a prism is $\sqrt{\frac{7}{3}}$ and the angle of prism is 60° . The minimum angle of incidence of a ray that will be transmitted through the prism is :

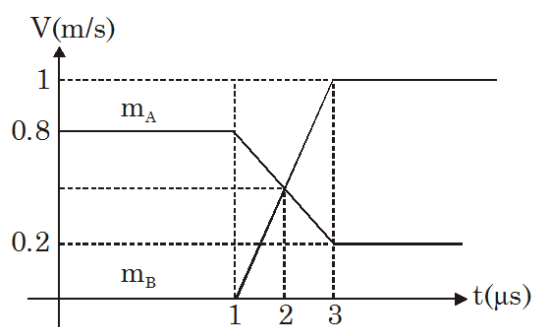
(A) 30° (B) 45° (C) 15° (D) 50°

12. A positive point charge Q is placed (on the axis of disc) at a distance of $4R$ above the centre of a disc of radius R . The magnitude of electric flux through the disc is ϕ . Now a hemispherical shell of radius R is placed over the disc such that it forms a closed surface. The flux through the curved surface taking direction of area vector along outward normal as positive, is :



(A) Zero (B) ϕ (C) $-\phi$ (D) 2ϕ

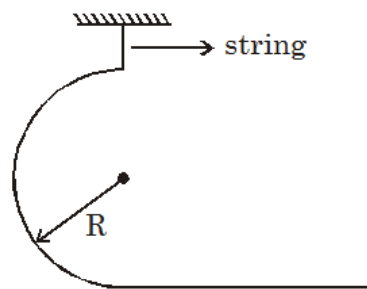
13. There are two blocks A and B placed on a smooth surface. Block A has mass 10 kg and it is moving with velocity 0.8 m/s towards stationary block B of unknown mass. At the time of collision, their velocities are given by the following graph. Coefficient of restitution of the collision is :



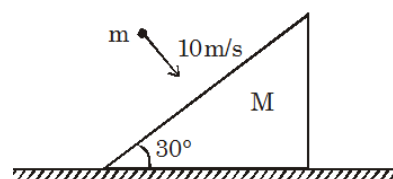
- (A) 1 (B) 1.5 (C) 0.5 (D) 0.8
14. A thin equiconvex lens, made of material of refractive index 1.5, has a focal length of 10 cm in air. One side of the lens is replaced by a medium (B) of refractive index 1.3. When a point object is placed in air at a distance of 20 cm from the lens on its principal axis, its image is formed at a distance of x cm, in the medium (B) from the lens. When the same object is placed in medium (B), at a distance of 20 cm from the lens on its principal axis, its image is formed at a distance of y cm in the air, from the lens then :

- (A) $x = y$ (B) $x = 1.3 y$
(C) $x > 1.3 y$ (D) $x < 1.3 y$

15. An object having linear mass density is suspended in a vertical plane in equilibrium as shown in the figure. The value of $\frac{L}{R}$ is :

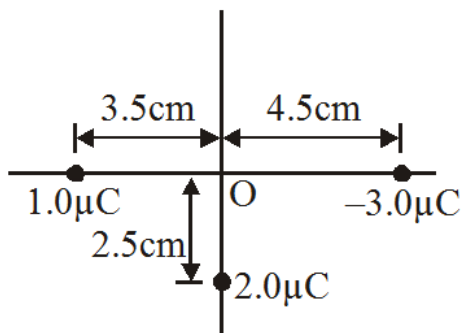


- (A) $\frac{1}{2}$ (B) π (C) 2 (D) $\frac{1}{2\pi}$
16. A simple pendulum is hanged inside a lift. It has a time period T_1 when lift was moving with constant velocity, and when the lift moves vertically up according to equation $y = 2t^2$, (where t is time in second) it has a time period T_2 , then $\left(\frac{T_2}{T_1}\right)^2$.
- (A) $\frac{6}{5}$ (B) $\frac{5}{7}$ (C) $\frac{7}{5}$ (D) $\frac{5}{6}$
17. A ball of mass 1 kg moving with velocity 10 m/s collides perpendicularly on a smooth stationary wedge of mass 2 kg. If the coefficient of restitution is $e = \frac{7}{20}$, then the velocity of ball after collision is



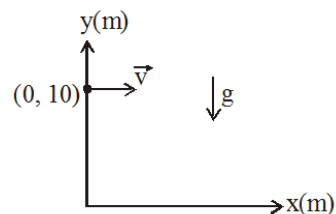
- (A) 5 m/s (B) $\frac{7}{2}$ m/s
(C) 2 m/s (D) 8 m/s

18. Three charges $1.00 \mu\text{C}$, $2.00 \mu\text{C}$ and $-3.00 \mu\text{C}$ are arranged as shown in the figure.

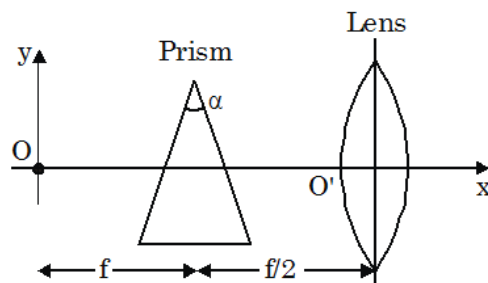


- (A) Total electric flux that leave a sphere centered at the origin and having a radius 2.00 cm is $2.2 \times 10^5 \text{ Nm}^2\text{C}^{-1}$
- (B) Total electric flux that leave a sphere centered at the origin and having a radius 3.00 cm is approximately $1.1 \times 10^5 \text{ Nm}^2\text{C}^{-1}$
- (C) Total electric flux that leave a sphere centered at the origin and having a radius 4.00 cm is approximately $3.3 \times 10^5 \text{ Nm}^2\text{C}^{-1}$
- (D) Total electric flux that leave a sphere centered at the origin and having a radius 5.00 cm is $2.2 \times 10^5 \text{ Nm}^2\text{C}^{-1}$

19. A charged particle of mass $m = 1 \text{ kg}$ & charge $Q = 1 \text{ C}$, is projected from $(0, 10)$ with velocity $\vec{v}$. Uniform electric field $(-10\hat{i} + 5\hat{j}) \text{ N/C}$ exist in the region. Gravity act along negative y direction. (take : $g = 10 \text{ m/s}^2$) For $\vec{v} = 10\hat{i}$



- (A) Particle never passes through origin.
- (B) Time spent by charged particle in I quadrant is 4 seconds.
- (C) While crossing origin, the trajectory makes equal angle with both x and y axes.
- (D) K.E. of particle at $(0, 10)$ will be equal to the K.E. at origin.
20. In the system shown in the diagram, rays from a point object are first deviated by a prism and then focused by a thin lens of focal length f . The prism angle is small (α) and its refractive index is n .



If the distance between object, prism and lens are as shown in figure, find the coordinate of image. Let origin is at the optical center of lens (O').

- (A) $[f, -f(n-1)\alpha]$ (B) $[2f, -f(n-1)\alpha]$
- (C) $[2f, -f(n-1)\alpha]$ (D) $[3f, -2f(n-1)\alpha]$

SEC-B : (Maximum Marks: 40)

This section contains 10 questions. Candidates have to attempt any 5 questions out of 10. If more than 5 questions are attempted, then only first 5 attempted questions will be evaluated.

The answer to each question is a Numerical Value Type question.

For each question, enter the correct numerical value (in decimal notation, truncated/rounded off to the second decimal place; e.g. 6.25, 7.00, -0.33, -0.30, 30.27, -127.30, if answer is 11.36777..... then both 11.36 and 11.37 will be correct)

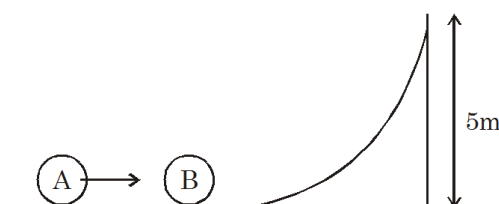
Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +4 If ONLY the correct numerical value is entered as answer.

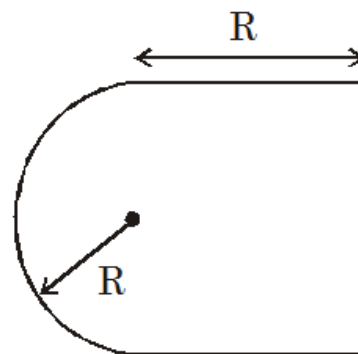
Zero Marks : 0 In all other cases.

- Electrostatic potential in a region is given by $V = 3x + 4y$ volts (x and y are in meter). The electrostatic potential energy stored in the cube of side 2m (in Joules) is $n\epsilon_0$. The value of n is :
- The objective of an astronomical telescope has a focal length of 80 cm . The telescope is focused for normal vision of distant objects when the tube length is 1.0 m . If the focal length of eye piece is f_e (in cm) and magnifying power of telescope is m then the value of $\frac{f_e}{m}$ is :

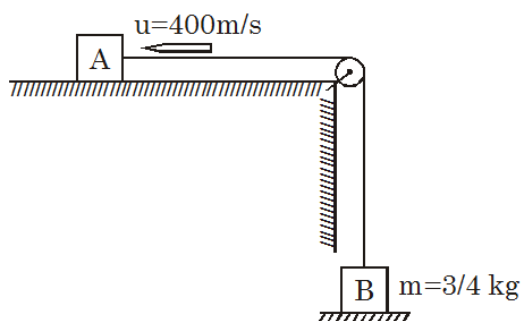
- Two balls A and B of masses m and $2m$ respectively are lying on a smooth surface as shown in the figure. Ball A hits the ball B (which is at rest) with a velocity $v = 16\text{ m/s}$, such that B just reaches highest point of inclined plane. If e is the coefficient of restitution and speed of A after collision is V_A , then the value of eV_A is :



- A small object placed in air at 15 cm from convex surface of a rod of glass ($n = 1.6$) of length R has a convex hemispherical surface of radius of curvature $R = 5\text{ cm}$ and a flat surface as shown in figure. Find the final distance of image (in cm) from the plane surface.

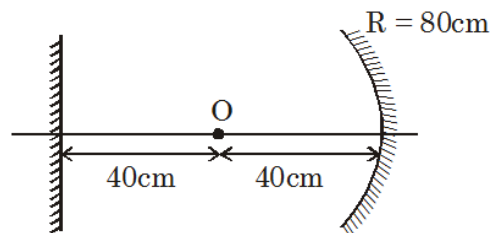


5. A block 'A' of mass $M_A = 1 \text{ kg}$ is kept on a frictionless horizontal surface and attached by a light thread to another block 'B' of mass $M_B = 0.75 \text{ kg}$. Block 'B' is resting on the ground and thread and pulleys are massless and frictionless. A small bullet of mass $m = 0.25 \text{ kg}$ moving with speed $u = 400 \text{ m/s}$ horizontally towards left sticks to the block 'A' as shown in the figure. Speed of block 'B' (in m/s) just after collision between the block 'A' and bullet is x . Then the value of x is :

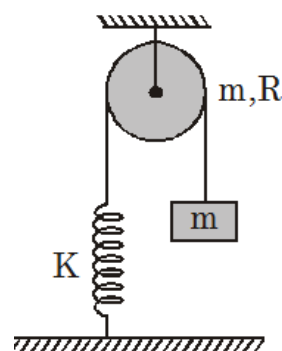


6. A ring of radius 0.9 m carries a total charge 0.1 nC distributed uniformly on it. E is electric field on the axis of ring at a distance x from centre. Find the value of $\int_{x=0}^{x=\infty} E \, dx$ in volt.
7. Two thin symmetrical lenses of different nature have equal radii of curvature ($R = 20 \text{ cm}$) of all faces. The lenses are kept in contact and immersed in water ($\mu = 4/3$). The difference in the refractive indices of the two lenses is $\frac{5}{9}$. The magnitude of focal length of the system is f in cm. Find the value of f .

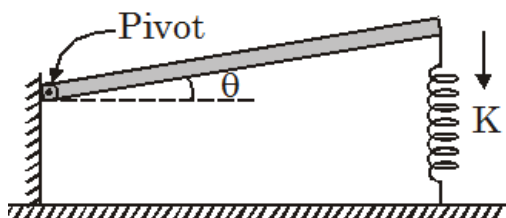
8. If direct rays only strikes on plane mirror, distance of first image formed by concave mirror is $\lambda \text{ cm}$ from concave mirror, then find λ .



9. A uniform disc of mass m and radius R is free to rotate about its fixed horizontal axis without friction. There is sufficient friction between the inextensible light string and disc to prevent slipping of string over disc. At the shown instant extension in light spring is $\frac{3mg}{K}$, where m is mass of block, g is acceleration due to gravity and K is spring constant. Then at the shown moment, magnitude of acceleration of block is xg (in m/s^2). Then value of x is :-



10. A horizontal rod of mass m and length L is pivoted smoothly at one end. The other end of rod is supported by a spring of force constant k . The rod is rotated (in vertical plane) by a small angle θ from its horizontal equilibrium position and released. The angular frequency of the subsequent simple harmonic motion is $\sqrt{\frac{PK}{m}}$. Then value of P is :



Topic :

CHEMISTRY

SEC-A : (Maximum Marks: 80)

This section contains **20 questions**. Each question has 4 options for correct answer. Multiple-Choice Questions (MCQs) **Only one option is correct**. For each question, marks will be awarded as follows:

Full Marks : +4 If correct answer is selected.

Zero Marks : 0 If none of the option is selected.

Negative Marks : -1 If wrong option is selected.

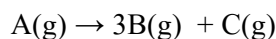
1. Which of the following species contains 3c-2e bonds and non planar geometry.

(A) Be_2H_4 (B) B_2H_6
(C) Al_2Cl_6 (D) C_2H_6

2. Which of the following compound is more acidic

(A) CH_4
(B) $\text{CH}_3\text{SO}_3\text{H}$
(C) $\text{CH}_3\text{CO}_2\text{H}$
(D) CH_3OH

3. At 100°C gaseous reaction



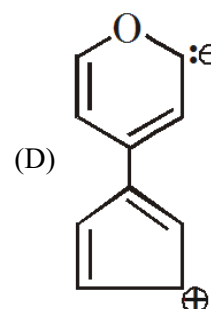
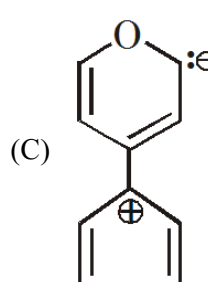
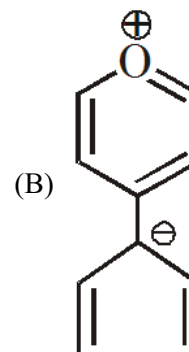
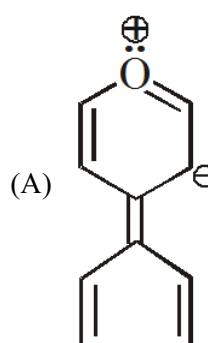
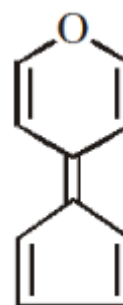
follows I^{st} order kinetics. After 10 mins total pressure of system is 325 mm Hg and after a long time total pressure is 400 mm Hg. Then half life of reaction is-

(A) 5 min.
(B) 10 min.
(C) 20 min.
(D) 2.5 min.

4. If we consider no mixing of '2s' and '2p' orbitals, then the bond order and magnetic nature of the diatomic molecule C_2 is –

(A) 3 and diamagnetic
(B) 2.5 and diamagnetic
(C) 2 and diamagnetic
(D) 2 and paramagnetic

5. The most stable resonating structure for the given molecule :



6. A transition metal M can exist in two oxidation states +2 and +3. It forms an oxide whose experimental formula is given by M_xO where $x < 1$. Then the ratio of metal ions in +3 state to those in +2 state is given by :

- (A) $\frac{(1-x)}{(1+x)}$ (B) $1+2x$
 (C) $1+\frac{x}{2}$ (D) $\frac{2(1-x)}{(3x-2)}$

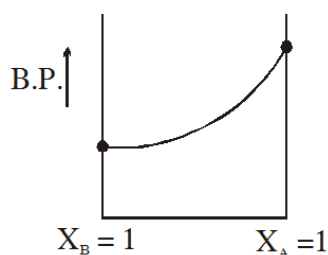
7. Trigonal pyramidal geometry around the underlined atom is shown by :

- (A) $\underline{N}(\text{SiH}_3)_3$ (B) $\underline{P}(\text{SiH}_3)_3$
 (C) $\underline{N}(\text{CH}_3)_3$ (D) Both (B) and (C)

8. Which conformation of 2-Fluoroethanol is most stable ?

- (A) Fully eclipsed (B) Partially eclipsed
 (C) Gauche (D) Anti

9. Based on below graph plotted between boiling point and mole fraction in liquid phase for the liquid binary mixture. Comment on the liquid mixture.

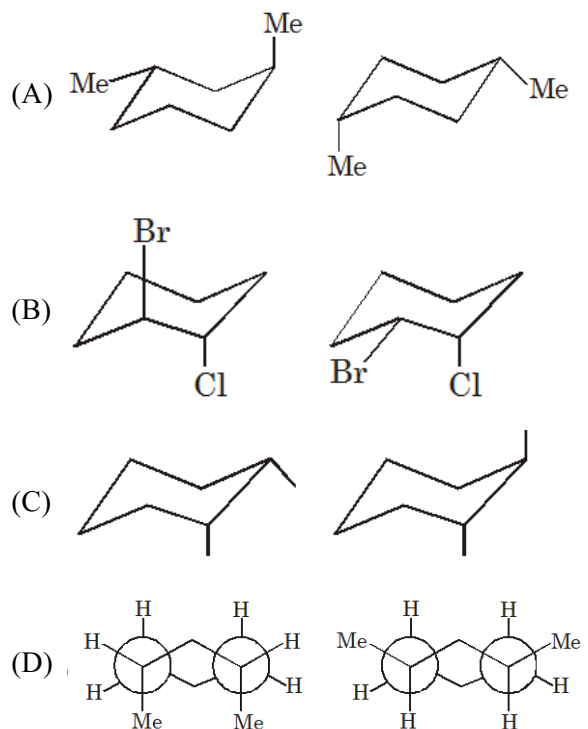


- (A) It is an ideal solution
 (B) Showing negative deviation from Raoult's law with formation of azeotrope
 (C) Showing positive deviation from Raoult's law with formation of azeotrope
 (D) Mixture of two immiscible volatile liquids

10. Which is **NOT CORRECT** for covalent character?

- (A) $\text{SbCl}_5 > \text{SbCl}_3$ (B) $\text{CaCl}_2 > \text{ZnCl}_2$
 (C) $\text{SnCl}_4 > \text{SnCl}_2$ (D) $\text{HgCl}_2 > \text{ZnCl}_2$

11. Which one is related as conformer among the given option :



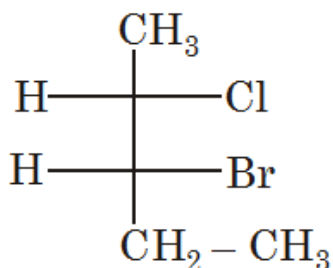
12. An azeotropic solution of two liquids has a boiling point lower than either of them when it :

- (A) Shows negative deviation from Raoult's law
 (B) Shows positive deviation from Raoult's law
 (C) Shows ideal behaviour
 (D) Is saturated

13. XeF_6 on complete hydrolysis gives :

- (A) Xe (B) XeO_2
 (C) XeO_3 (D) XeO_4

14. The correct nomenclature is :



- (A) 2R, 3R
(B) 2R, 3S
(C) 2S, 3R
(D) 2S, 3S

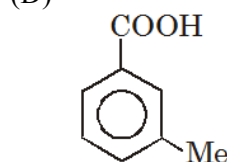
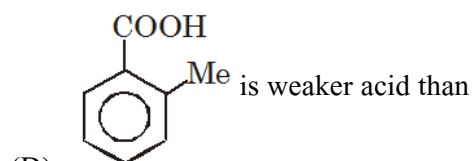
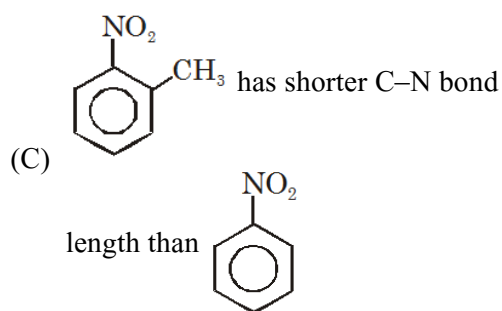
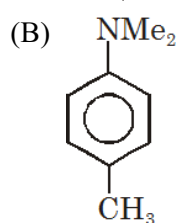
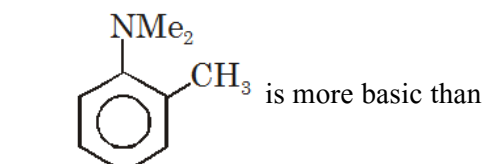
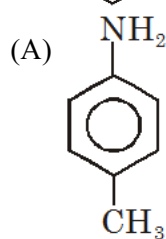
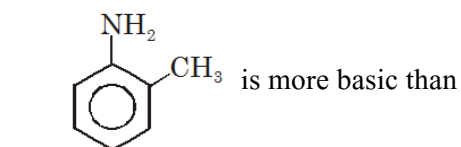
15. Which of the following is the incorrect for critical temperature -

- (A) It is the highest temperature at which liquid and vapour can coexist
(B) Beyond the critical temperature, gas cannot be liquefied by compression
(C) Higher is the critical temperature easier to liquefy the gas
(D) At critical temperature, the gas behaves like ideal gas.

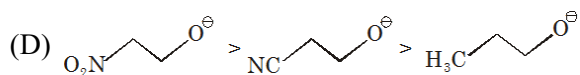
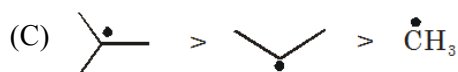
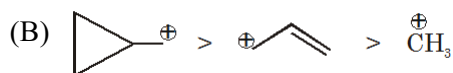
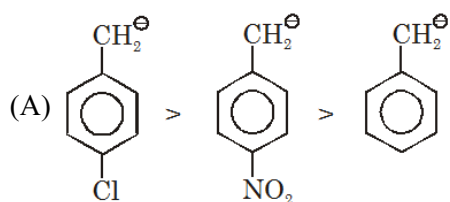
16. Number of P-O-P bonds present in pentamer form of cyclometaphosphoric acid.

- (A) 3
(B) 5
(C) 7
(D) 10

17. Select the correct statement(s) ?



18. The vapour pressure of an aqueous solution of glucose is 750 mm of Hg at 373 K. The molality is :
- (A) 0.64 (B) 0.741
(C) 0.68 (D) 0.94
19. Which of the following species is paramagnetic with only one unpaired electron, which is present in bonding molecular orbital.
- (A) C_2^{2-} (B) O_2^+
(C) O_2^- (D) N_2^+
20. Identify incorrect stability order for following intermediates / ions ?



SEC-B : (Maximum Marks: 40)

This section contains 10 questions Candidates have to attempt any 5 questions out of 10. If more than 5 questions are attempted, then only first 5 attempted questions will be evaluated.

The answer to each question is a Numerical Value Type questions.

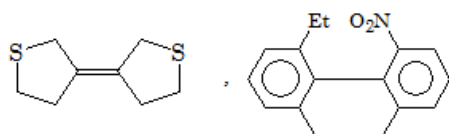
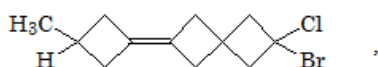
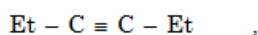
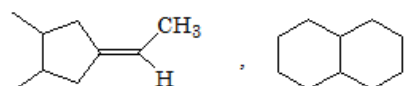
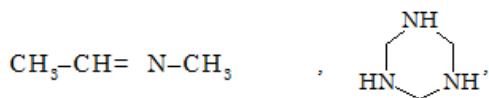
For each question, enter the correct numerical value (in decimal notation, truncated/rounded off to the second decimal place; e.g. 6.25, 7.00, -0.33, -0.30, 30.27, -127.30, if answer is 11.36777..... then both 11.36 and 11.37 will be correct)

Answer to each question will be evaluated according to the following marking scheme:

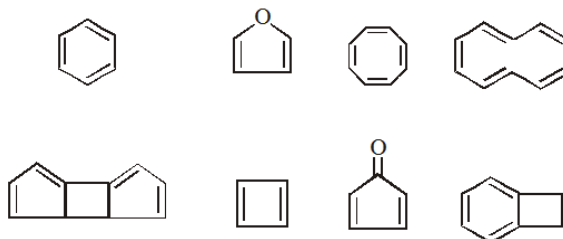
Full Marks : +4 If ONLY the correct numerical value is entered as answer.

Zero Marks : 0 In all other cases.

1. If in a simple cubic arrangement 'X' number of faces meet at a corner of a cube in crystal lattice then what is the value of (X-3)
2. Find the number of molecules which gives H_2SO_4 as hydrolysis product or one of the hydrolysis product.
 SF_4 , $\text{H}_2\text{S}_2\text{O}_7$, $\text{H}_2\text{S}_2\text{O}_8$, SO_3 , SOF_2 , POCl_3
3. Find the number of compound which can show geometrical isomerism



4. The time taken for a certain volume of gas to diffuse through a small hole was 2 min. Under similar conditions an equal volume of oxygen took 5.65 min to pass. The molecular mass of the gas in amu is (use $4\sqrt{2} = 5.65$) :
5. The total number of species which have bond order greater than two
 O_2 , O_2^+ , O_2^- , N_2 , N_2^+ , CN^- , B_2 , C_2
6. Identify aromatic compound ?



7. $k_1 = 10^{-2} \text{ M sec}^{-1}$
 $k_2 = 2 \times 10^{-2} \text{ M sec}^{-1}$
Find concentration of 'C' (mol/L) at 100 sec.
8. Find out the number of planar species
 ClF_3 , I_3^+ , I_3^- , XeF_4 , XeF_5^- , I_2Cl_6 , SO_2Cl_2 , SF_4
9. Find number of compound which can show Tautomerism
10. If NaCl is doped with 10^{-3} mol % of AlCl_3 what is the number of cationic vacancies per mole of NaCl. (Given $N_A = 6 \times 10^{23}$)
If your answer in scientific notation is $x \times 10^y$ then fill x in OMR Sheet.

Topic :

MATHEMATICS

SEC-A : (Maximum Marks: 80)

This section contains **20 questions**. Each question has 4 options for correct answer. Multiple-Choice Questions (MCQs) **Only one option is correct**. For each question, marks will be awarded as follows:

Full Marks : +4 If correct answer is selected.

Zero Marks : 0 If none of the option is selected.

Negative Marks : -1 If wrong option is selected.

- The range of $f(x) = \frac{x}{1+x^2}$ is
(A) $[0, \infty)$ (B) $\left[-\frac{1}{2}, \frac{1}{2}\right]$
(C) $[0, 1]$ (D) $(-\infty, \infty)$
- Domain of the function $f(x) = \frac{3}{4-x^2} + \log_{10}(x^3 - x)$ is
(A) $(-1, 0) \cup (1, 2)$
(B) $(1, 2) \cup (2, \infty)$
(C) $(-1, 0) \cup (1, 2) \cup (2, \infty)$
(D) $(1, 2)$
- $\lim_{x \rightarrow 1} \left(\sec\left(\frac{\pi x}{2}\right) \log x \right)$ is
(A) $-\frac{2}{\pi}$ (B) $-\frac{\pi}{2}$ (C) $\frac{2}{\pi}$ (D) $\frac{\pi}{2}$
- $\lim_{x \rightarrow \frac{\pi}{6}} \frac{3 \sin x - \sqrt{3} \cos x}{6x - \pi} =$
(A) $\sqrt{3}$ (B) $\frac{1}{\sqrt{3}}$ (C) $-\sqrt{3}$ (D) $-\frac{1}{\sqrt{3}}$

- The values of a, b and c such that $\lim_{x \rightarrow 0} \frac{axe^x - b \log(1+x) + cxe^{-x}}{x^2 \sin x} = 2$
(A) $a = 3, b = 12, c = 9$
(B) $a = 1, b = 2, c = 4$
(C) $a = 2, b = 10, c = 8$
(D) $a = 3, b = -12, c = -9$
- If $\ell_1 = \lim_{x \rightarrow 2^+} (x + [x])$, $\ell_2 = \lim_{x \rightarrow 2^-} (2x - [x])$ and $\ell_3 = \lim_{x \rightarrow \frac{\pi}{2}} \frac{\cos x}{x - \frac{\pi}{2}}$, then [where [] denotes G.I.F.]
(A) $\ell_1 < \ell_2 < \ell_3$ (B) $\ell_2 < \ell_3 < \ell_1$
(C) $\ell_3 < \ell_2 < \ell_1$ (D) $\ell_1 < \ell_3 < \ell_2$
- If $0 < x < 1$ then $\tan^{-1} \left(\frac{\sqrt{1-x^2}}{1+x} \right)$ is equal to
(A) $\frac{1}{2} \cos^{-1} x$ (B) $\cos^{-1} \sqrt{\frac{1+x}{2}}$
(C) $\sin^{-1} \sqrt{\frac{1-x}{2}}$ (D) All the above
- If $n \in \mathbb{N}$, $\sum_{k=1}^n \sin^{-1}(x_k) = \frac{n\pi}{2}$, then $\sum_{k=1}^n x_k =$
(A) n (B) k
(C) $\frac{k(k+1)}{2}$ (D) $\frac{n(n+1)}{2}$
- If $x(3-x) \geq 2$ then $\sin^{-1}(x) + \sin^{-1}(x^2) + \dots + \sin^{-1}(x^{10}) =$
(A) $\frac{\pi}{2}$ (B) 2π (C) 5π (D) 10π

10. The function

$$f(x) = \begin{cases} \frac{\cos 3x - \cos 4x}{x^2}, & \text{for } x \neq 0 \\ \frac{7}{2}, & \text{for } x = 0 \end{cases} \text{ at } x = 0 \text{ is}$$

- (A) continuous
(B) discontinuous
(C) left continuous
(D) right continuous

11. Let

$$f(x) = \begin{cases} \frac{(e^x - 1)^{2n}}{\sin^n(x/a)(\log(1+(x/a)))^n}, & \text{for } x \neq 0 \\ 16^n, & \text{for } x = 0 \end{cases}$$

and f is a continuous at $x = 0$, then the value of a is

- (A) 16 (B) 2
(C) 8 (D) 4

12. Let $f(x) = x^{3/2} - \sqrt{x^3 + x^2}$ then

- (A) $f(x)$ is non differentiable at $x = 2$
(B) $f(x)$ is discontinuous at $x = 5$
(C) $f(x)$ is discontinuous and non differentiable at $x = 1$
(D) $f(x)$ is continuous and differentiable at $x = 0$

13. The set of all points where the function

$$f(x) = \sqrt{1 - e^{-x^2}}$$

- is differentiable is
(A) $(0, \infty)$
(B) $(-\infty, \infty)$
(C) $(-\infty, \infty) - \{0\}$
(D) $(-\infty, \infty) - \{0, 1, 2\}$

14. If $0 < a < 1$ and $f(x)$ is defined as $f(x) = \begin{cases} x^a & \text{if } x > 0 \\ 0 & \text{if } x = 0 \end{cases}$ then

- (A) $f(x)$ is differentiable at $x = 0$
(B) $f(x)$ is not continuous
(C) $f(x)$ is not differentiable at $x = 0$
(D) $f(x)$ is bounded

15. Suppose for a differentiable function f , $f(0)=0$, $f(1) = 1$ and $f'(0) = 4 = f'(1)$. If $g(x) = f(e^x)$. $e^{f(x)}$ then $g'(0)$ is

- (A) 4 (B) 8 (C) 2 (D) -2

16. Equation of the locus of all points such that the absolute difference of its distances from $(-3, -7)$, $(-3, 3)$ is 8 is

- (A) $\frac{(x+3)^2}{16} - \frac{(y+2)^2}{9} = 1$
(B) $\frac{(x+3)^2}{9} - \frac{(y+2)^2}{16} = -1$
(C) $\frac{(x+3)^2}{9} - \frac{(y+2)^2}{19} = 1$
(D) $\frac{(x+3)^2}{7} - \frac{(y+2)^2}{19} = -1$

17. A hyperbola, having the transverse axis of length $2\sin\theta$, is confocal with the ellipse $3x^2 + 4y^2 = 12$. Then its equation is

- (A) $x^2 \operatorname{cosec}^2\theta - y^2 \sec^2\theta = 1$
(B) $x^2 \sec^2\theta - y^2 \operatorname{cosec}^2\theta = 1$
(C) $x^2 \sin^2\theta - y^2 \cos^2\theta = 1$
(D) $x^2 \cos^2\theta - y^2 \sin^2\theta = 1$

18. Consider the parabola $y^2 = 4x$; A(4, -4); B(9,6) be two fixed points on the parabola. Let 'C' be a moving point on the parabola between A and B such that the area of the triangle ABC is maximum. The coordiante of C is
- (A) $\left(\frac{1}{4}, 1\right)$ (B) (4,4)
 (C) $(3, 2\sqrt{3})$ (D) $(3, -2\sqrt{3})$
19. The tangent at any point P on the ellipse meets the tangents at the vertices A and A' of the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ at L and M respectively. Then AL.A'M =
- (A) a^2 (B) b^2 (C) $a^2 + b^2$ (D) ab
20. For the hyperbola $\frac{x^2}{25} - \frac{y^2}{16} = 1$ which of the following is not correct
- (A) length of latus rectum is equal to $\frac{32}{5}$
 (B) eccentricity is $\frac{\sqrt{41}}{5}$
 (C) centre is (0,0)
 (D) distance between focii is $2\sqrt{45}$

SEC-B : (Maximum Marks: 40)

This section contains 10 questions Candidates have to attempt any 5 questions out of 10. If more than 5 questions are attempted, then only first 5 attempted questions will be evaluated.

The answer to each question is a Numerical Value Type questions.

For each question, enter the correct numerical value (in decimal notation, truncated/rounded off to the second decimal place; e.g. 6.25, 7.00, -0.33, -30, 30.27, -127.30, if answer is 11.36777..... then both 11.36 and 11.37 will be correct)

Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +4 If ONLY the correct numerical value is entered as answer.

Zero Marks : 0 In all other cases.

1. $\lim_{x \rightarrow 0} \frac{1 - \cos 3x \cos 9x \cos 27x \dots \cos x (3^n x)}{1 - \cos \frac{x}{3} \cos \frac{x}{9} \cos \frac{x}{27} \dots \cos \frac{x}{3^n}} = 3^{10}$

where $n \in \mathbb{N}$ then $n =$

2. If $y = \sin^{-1} [x\sqrt{1-x} - \sqrt{x}\sqrt{1-x^2}] + \cos^{-1} x - \cos^{-1}(\sqrt{x})$ and $0 < x < 1$, then $\frac{dy}{dx}$ at $x = \frac{\sqrt{3}}{2}$

3. If the vertex of a hyperbola bisects the distance between its centre and the corresponding focus, then ratio of square of its conjugate axis to the square of its transverse axis is

4. Total number of tangents of the hyperbola $\frac{x^2}{9} - \frac{y^2}{4} = 1$, that are perpendicular to the line $5x + 2y - 3 = 0$ is

5. Radius of the largest circle which passes through the focus of the parabola $y^2 = 4x$ and contained in it, is

6. If $(\alpha, \alpha - 1)$ lies inside the ellipse $16x^2 + 9y^2 - 16x = 0$ such that complete set of values of α is $\left(\frac{9}{25}, k\right)$. The value of k is

7. If fractional part of $\frac{1}{x}$ and x^2 for some $x \in (\sqrt{2}, \sqrt{3})$ are equal, then the value of $x^4 - \frac{3}{x}$ is

8. Consider $f: \mathbb{R}^+ \rightarrow \mathbb{R}$ such that $f(3) = 1$ for $a \in \mathbb{R}^+$ and $f(x) \cdot f(y) + f\left(\frac{3}{x}\right) \cdot f\left(\frac{3}{y}\right) = 2f(xy)$ $\forall x, y \in \mathbb{R}^+$, then $f(97) =$

9. Let $f(x) = \begin{cases} \frac{1+\cos \pi x}{\pi^2(x+1)^2} & , x < -1 \\ ax + b & , -1 \leq x \leq 1 \\ \frac{x-1}{\ln \sqrt{x}} & , x > 1 \end{cases}$, f is

continuous $\forall x \in \mathbb{R}$, then value of $3a - b$ is

10. If $f(x) = (2x - 3\pi)^5 + \frac{4}{3}x + \cos x$ and g is the inverse function of f , then $g'(2\pi)$ is equal to

Space for Rough Work